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1) Explain Classical cryptography with suitable examples.

→ Classical Cryptography is an old method of securing information by converting readable text into unreadable form using an encryption technique and a key.

⇒ Substitution Cipher :→ In a substitution cipher, each letter of Plain text is replaced with another letter according to a specific rule or key.

Ex :→ Caser cipher with key = 3

Plaintext →	H	E	L	L	O
	↓	↓	↓	↓	↓
ciphertext :→	K	H	O	O	R

⇒ Transposition cipher :→ The positions of letters are changed but the letters remain the same.

Ex :→ Plaintext :→ TEAM
ciphertext :→ META

⇒ Goals :→ Confidentiality, Integrity, Availability.

⇒ Advantages :→ simple to understand and implement
:→ uses easy encryption technique.

⇒ Limitation :→ Easily broken using modern techinque
:→ Not suitable for modern security.

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2) Explain working of :-

1) Caser cipher :- Caser cipher is a subsitition technique in which each letter is shiffted by a fixed number of positions.

Ex :- Key :- 3
Plaintext :- HELLO
Ciphertext :- KHOOR

~~2) Vigenere cipher~~

for Encryption :- Plaintext + key
Decryption :- Ciphertext - key

=> Advantages :- Simple and easy to understand
:- fast encryption and Decryption

=> Disadvantages :- use only one fixed key.
:- Not secure for modern Communicatio

2) Vigenere Cipher :- vigenere cipher is a subsitition technique that uses a keyword to encrypt the Plaintext. Each letter is shifted according to the corresponding keyword letter.

Ex :- Plaintext :- NETWORKSECURITY
key :- ATTACK

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Plaintext :→	N	E	T	W	O	R	K	S	E	C	U	R	I	T	Y	
	13	4	19	22	14	17	10	18	4	2	20	17	8	19	24	
Key :→	A	T	T	A	C	K	A	T	T	A	C	K	A	T	T	
	0	19	19	0	2	10	0	19	19	0	2	10	0	19	19	
(P+K) mod 26 :→	13	23	38	22	16	27	10	37	23	2	22	27	8	38	43	
			(12)			(1)		(11)				(1)		(12)	(17)	
Cipher :→	N	X	M	W	Q	B	K	L	X	C	W	B	I	M	R	

Output :→ Plaintex :→ NETWORKSECURITY
key :→ ATTACKATTACKATT
Ciphertext :→ NXMWQBKLXCWBIMR

3) Playfair Cipher :→ Playfair Cipher encrypts letters in pair using 5x5 key matrix.
→ The letters I and J are usally combined.

Ex :→ Plaintext :→ CRYPTOGRAPHY
key :→ PPSAVANIUNIVERSITY

Unique letters from the key :→ PSAVNIVERTY

P S A V N
I U E R T
Y B C D F
G H K L M
O Q W X Z

Plaintext :→ CRYPTOGRAPHY
Ciphertext :→ BEGIIZLIVSGB

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3) Describe the CIA triad with suitable real world example.

→ The CIA triad is a basic model of network security that contains of three principals:

1) Confidentiality :→ Ensure that information is accessible only to authorized people.

Ex :→ only the account holder can access their bank account details.

2) Integrity :→ Ensure that ~~the~~ data is accurate and is not changed without permission.

Ex :→ A student's marks cannot be changed ~~without permission~~ by an unauthorized person.

3) Availability :→ Ensure that data and services are avilable whenever needed.

Ex :→ An ATM should be avilable for customers to withdraw money when required.

4) Compare Symmetric and Asymmetric Encryption Techniques with their advantages, Disadvantages and applications

I → Symmetric :→ In symmetric Encryption, the same secreate key is used for both encryption and decryption.

Example :→ AES, DES

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Advantages :→ Simple to implement
:→ Suitable for encrypting large files.

Disadvantages :→ Difficult to share the secrete key security.
:→ Does not provide digital signature.

II =→ Asymmetric Encryption :→ In Asymmetric Encryption, two diffrent keys are used:
- Public key :→ use for encryption
- Private key :→ use for discryption.

Advantages :→ Provides better security
:→ secure key exchange.

Disadvantages :→ slower than symmetric encryption
:→ more Complex to implement.

Ex:→ RAS

Symmetric	Asymmetric
→ Symmetric key cryptography only one key.	Asymmetric key cryptography use two key.
→ same key is used for encryption and dicryption	Diffrent keys are used for encryption and decryption.
→ secrent key must be shared security.	Public key can be shared openly. Privat key is keep secret
→ Speed Faster	Slower than symmetric crypto.
→ secure but secure key sharing is difficut.	Provides secure key exchange and authentication

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5) * Difference between DES, 3DES and AES.

	DES	3DES	AES
→	Data Encryption Standard	Triple Data Encryption Standard	Advance Encryption standard
→	Symmetric key Block chiphar	symmetric key Block ciphar	symmetric key Block ciphar
→	64 bits Block size	64 bits Block size	128 bits Block size
→	56 bits key size	112 or 168 bits key size	128, 192 or 256 bits key size
→	DES encryption once	DES applied three times	Substitution and Permutation based
→	Faster than 3DES	slow	Faster than 3DES & DES
→	Feistel Network structure	Feistal Network Structure	Substitution - Permutation Network structure
→	low security	Better than DES	very highy security
Ex:→	Data ↳DES ↳ciphertext	Data ↳DES->DES ↳DES ↳ciphertext	Data ↳AES ↳ciphertext

6) Explain the AES algorithm with neat diagram of it's structure.

→ AES (Advanced Encryption Standard) is a symmetric-key block cipher used for secure encryption and descryptio of data.

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→ It replaced DES because it provides stronger security and better performance.

→ AES work on 128-bit block of plaintext
→ It supports 128-bit, 192 bit and 256-bit keys.
→ AES has 10, 12 or 14 rounds, depending on the key size.
→ It use a subsitition-Permutation network structure.
→ The 128 bit data is arged in a 4x4 matrix of bytes, call the state.
→ AES is used in wifi, VPNs, file encryption and secure communication.

=> Advantages :→ AES gives strong protection for data.
:→ AES faster than DES and 3DES.

=> Disadvantages :→ key managment
:→ Implementation.

7) Find the GCD (750, 900)

=> Formula :→ $A = B \times Q + R$
where, A = large no.
B = small no.
Q = quotient
R = reminder

Q	A	B	R
1	900	750	150
5	750	150	0
	150	0	

∴ GCD (750, 900) = 150

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8> RAS :->

S:->1> Two Prime NO decide

P = 17 q = 11

S:->2> Find the value of n

∴ n = P × q

= 17 × 11

= 187

S:->3> Find ϕ(n)

∴ ϕ(n) = (P-1)(q-1)

= (17-1)(11-1)

ϕ(187) = 160

S:->4> Select the Public exponent e

choose e such that : 1 < e < ϕ(n) :

and

GCD (e, ϕ(n)) = 1

Here choose:

e = 7

check -

GCD (7, 160) = 1

Therefore e = 7

S:->5> Calculate the Private exponent d

d × e ≡ 1 (mod ϕ(n))

Substitute the value

d × 7 ≡ 1 (mod 160)

d = 23 beco2 (7×23) 161)

(161 - 1 = 160)

Therefore,

7 × 23 ≡ 1 (mod 160)

d = 23

S:->6> Generate RAS key

Public key (e, n) = (7, 187)

Private key (d, n) = (23, 187)

Answer:-

P = 17

q = 11

n = 187

ϕ(n) = 160

e = 7

d = 23

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9) Chinese Reminder theorm:

$x = 2 \pmod 3$

$x = 3 \pmod 5$

$x = 2 \pmod 7$

=> Given,

$a_1 = 2$ $\quad m_1 = 3$

$a_2 = 3$ $\quad m_2 = 5$

$a_3 = 2$ $\quad m_3 = 7$

S1) Find M,

$\therefore M = m_1 \times m_2 \times m_3$

$= 3 \times 5 \times 7$

$M = 105$

S2) Find m_1, m_2 & m_3

$M_1 = M/m_1 = 105/3 = 35$

$M_2 = M/m_2 = 105/5 = 21$

$M_3 = M/m_3 = 105/7 = 15$

S3) Find $m_1^{-1}, m_2^{-1}, m_3^{-1}$

$m_1 \times m_1^{-1} = 1 \bmod m_1$	$m_2 \times m_2^{-1} = 1 \bmod m_2$	$m_3 \times m_3^{-1} = 1 \bmod m_3$
$35 \times m_1^{-1} = 1 \bmod 3$	$21 \times m_2^{-1} = 1 \bmod 5$	$15 \times m_3^{-1} = 1 \bmod 7$
$35 \times 2 = 1 \bmod 3$	$21 \times 1 = 1 \bmod 5$	$15 \times 1 = 1 \bmod 7$
$m_1^{-1} = 2$	$m_2^{-1} = 1$	$m_3^{-1} = 1$

S4) APPLY CRT Foormula

$\therefore x = (a_1 m_1 m_1^{-1} + a_2 m_2 m_2^{-1} + a_3 m_3 m_3^{-1}) \bmod m$

$x = (2 \times 35 \times 2 + 3 \times 21 \times 1 + 2 \times 15 \times 1) \bmod 105$

$x = (140 + 63 + 30) \bmod 105$

$x = 233 \pmod{105}$

$\boxed{x = 23}$

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10) Explain network security:

→ Network security is the Process of Protecting a Computer network from unauthorized access, attacks, misuse and data theft.

→ It ensures the safety and Privacy of data during Communication.

1) Confidentiality :→ Ensures that only authorized users can access the data.

2) Integrity :→ Protects data from Unauthorized changes or modification.

3) Authentication :→ verifies the identity of users before allowing access to the network.

4) Access Control :→ Allow only authorized users to access specific network resources.

5) Firewall :→ Monitors incoming and outcoming network traffic and blocks unauthorized access.

6) Encryption :→ Converts readable data into an unreadable format to Protect it from attackers.

=> Tools :→ Firewall
Encryption
IDS
VPN

11) Explain network Scanning tools.

→ Network Scanning tools are software tools used to identify active devices, open Ports, running services and Possible security weaknesses in a Computer network.

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1) ~~Num~~Nmap :→ Network mapping is used to discover devices, scan open ports and identify running services on a network.

2) Angry IP scanner :→ Quickly scans IP addresses to find active devices on a network.

3) Nessus :→ Identifies security vulnerability and misconfiguration in system and network.

4) Netcat :→ used to test network connections and communicate through TCP or UDP ports.

5) Uses :→ These tools help network administrators find security problems, troubleshoot and improve network security.

12) Explain IDS and IPS - Differentiate between IDS and IPS.

⇒ IDS :→ Intrusion Detection System

:→ IDS is a security system that monitors network traffic and detects suspicious activities or attacks

:→ It generates alerts when a threat is found.

⇒ IPS :→ Intrusion Prevention System

:→ IPS is a security system that monitors network traffic and detects threats and automatically block malicious activites.

Ex:→ IPS :→ Detects the attack and blocks the malicious traffic

IDS :→ Detects the attack and sends an alerts to the administor.

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IDS	IPS
→ Intrusion Detection System	Intrusion Prevention System
→ Detects attacks	Detects and Prevent attack
→ Generates alerts	Automatically blocks threats
→ Usally works Passively	Usally works inline
→ Doesnot diretly block traffic	Can block malicious traffic

13) Explain shannon's Principles of Confusion and diffusion.

→ In 1949, claude shannon introduced two principles for designing secure cryptographic system:

=> Confusion :→ Confusion hides the relationship between the secret key and ciphertext.
:→ It makes it difficult for attackers to find the secret key.
:→ It is commonly achieved using substitution techniques.

=> Diffusion :→ Diffusion spreads the influence of plaintext bits over many ciphertext bits.
:→ It hides the statistical patterns of the original message.
:→ It commonly achieved using permutation and mixing techniques.

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⇒ Importance of Both Principals:→

- Confusion and diffusion make encrypted data more secure.
- They help Protect against statistical attacks.

⇒ Difference.

Confusion	Diffusion
→ Hides the relationship between key and ciphertext	Hides Patterns in the plaintext.
→ use substitution	Use permutation and mixing.
→ Makes key discovery difficult.	Make statsical analysis difficut.
Ex:→ AES subbytes.	AES shiftrow and Mixcolum.

14) Explain the DES Algorithm and feistel structure with a neat Block diagram:

→ DES (Data Encryption standard)

→ DES is a symmetric key encryption algorithm that ~~encrypt~~ encryptes 64-bit data blocks using a 56-bits effective key.

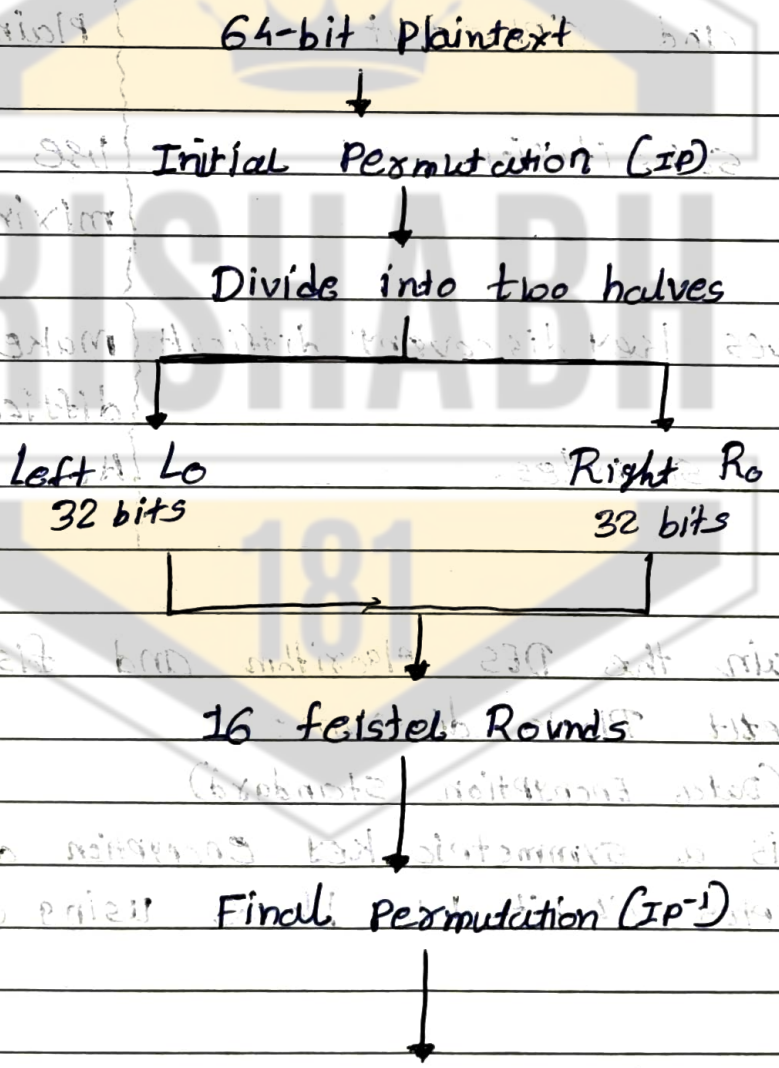
→ It uses the feistel Structure and Perform 16 round of encryption.

⇒ Input → DES takes 64 bits Plaintext block as input.

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=> IP :-> The input rearranged according to a fixed Permutation
=> Division :-> Splits the block into two 32-bit halves (Left and Right)
=> 16 Rounds :-> Performs expansion, XOR, substitution and permutation.
=> Final Permutation :-> Produces the 64-bit ciphertext.
=> key generation :-> Generates 16 round keys of 48 bits from a 56-bit effective key.

=> Diagram :->



=> Feistal Structure :->
-> The input block is divided into left and Right halves.
-> In each round, the right half is Processed using a

round function F and a round key k_i

→ The output is XORed with the left half.

→ The halves are swapped for the next round.

⇒ Equations:

$$L_i = R_{i-1}$$

$$\therefore R_i = L_{i-1} \oplus F(R_{i-1}, k_i)$$

⇒ AES Diagram :-

